MIELA MAYER · ESSAY

Learning to Let Nature Do the Work

A Review of Permaculture's Potential within the American Agriculture Industry

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ABSTRACT

Permaculture is an alternative agroecological movement that presents promising potential to improve our current system of food production while simultaneously reducing the agricultural repercussions that America faces. Although the practice's ideals and principals are gaining an increasing amount of public attention, the concept is far less prevalent within scientific research and literature. Thus, permaculture is widely used internationally but does not always have enough evidence to back its adaption into more regulated American agriculture. It is, however, very easy to adopt and follow the guidelines of permaculture on a small scale or as a private practice. Some recent initiatives are documenting their positive findings, even in temperate zones of the east coast. The following examples are further discussed in the text: University of Massachusetts Amherst's on-campus permaculture start up, Ben Falk's Whole System Design research farm in central Vermont, and the Copia farm in Johnstown, OH, supplemented with evidence from more extensive international examples in Jordan and New Zealand. Such international farms are useful for guidance and potentially for more intensive research and observation in the future since they have been established for much longer, and often on a large production scale. Permaculture's impending success lies in its broad framework, which integrates wisdom from long-practiced traditional and alternative land use, with modern scientific knowledge and research advancements. Further collaboration between these two fields of practice and analysis is needed to gain more institutional recognition of permacultures sustainable techniques and enable its implementation in America's large-scale agriculture where its benefits would have the most impact. This paper presents permaculture and its constituent realms of practice; primarily focusing on the more modern subset of agroforestry as it has the most documented results and a particularly promising potential to lower our carbon emissions and mitigate climate change.

THE PROBLEM WITH INDUSTRIAL AGRICULTURE

Most Americans are unaware of the problems inherent in industrialized agriculture, however, the sector's heavy dependence on external mechanized and chemical inputs is surely exhausting the planets finite supply of fossil fuels, arable cropland, and uncontaminated water, while directly contributing to greenhouse gas emissions and therefore overall climate change. According to the 2013 United Nations trade and environment review, Agriculture, land use changes, and forestry account for an accumulative 71 percent of all green house gas emissions. Not only is this monoculture-based system often growing genetically modified crops, unsustainable and environmentally damaging, but it is also now failing to provide the originally promised trade off for all its shortcomings: efficient and ample enough food production to keep up with the growing population's needs. Today with 7 billion people looking to 5 billion hectares of farmland for their food, crop yields are only increasing at a rate of half a percent per year. With the world's population expected to hit 9 billion people in 2050, this rate would need to be more than quadrupled in order to come anywhere near reaching the estimated over 50% increase in production that will be required to meet rising demands. On the bright side, while using the same science and technology that created "conventional farming techniques" people have also been able to design highly efficient, regenerative, and sustainable agriculture systems that are less reliant on fossil fuels and sequester more carbon. These methods, all of which fall under the overarching practice of permaculture, maintain healthy soil and support long-term productivity without synthetic chemical pesticides or fertilizers, thereby reducing petrochemical runoff into our rivers while using less water more efficiently.

WHAT PERMACULTURE IS

Similar to the green revolution's aim in shifting to industrial agriculture, permaculture serves as a highly productive form of agriculture, requiring little external labor. The difference is that rather than working against natural processes and destroying the environment's pre-existing polycultures, permaculture aims to constructively build upon such ecosystems, using the multipurpose diversity of species to construct a symbiotic, interdependent, and perennial supply of food. The resulting resiliency of these sustainable methods, comparable to that of a natural forest, enable permaculture to be applied and thrive virtually anywhere, unlike industrial farms. It is adaptable to a wide range of conditions on both small and large scales. This means that over time wide spread small scale permaculture implementation is not only feasible but also provides the best means of reversing environmental degradation, preventing climate change through carbon sequestration, and reducing Americas heavy reliance upon industrial agribusinesses.

In the 1970's Bill Mollison and David Holmgren, often deemed the fathers of permaculture, first defined the term to literally mean "permanent agriculture". Although this word has been interpreted and altered in many ways overtime and now has numerous subsets, for the purposes of this paper I will define modern permaculture as a broad field of sustainable agriculture that encompasses traditional farming practices like Hugelkultur, Keyline, and Coppice as well as the more recent practice of Agroforestry. To clarify what "sustainable agriculture" entails, we will use Gordon R Conway's environmentalist definition: "Sustainability means a responsibility for the environment—a stewardship of our natural resources. It means sufficient food and fiber that complements and, indeed, enhances our natural resource endowment of forests, soils and wildlife". Our tendency is to hold back the environments succession though weeding, mowing, pruning, etc. This "succession" however, is natures sustainable way of cycling minerals and organic mater to efficiently produce food while maintaining soil ecology, promoting biotic diversity, preventing erosion, and providing habitats for other animal species—all without high labor inputs and fossil fuel consumption.

AGROFORESTRY AND THE EUROPEAN YIELD COMPARISON

Agroforestry, a subset of permaculture, aims to mimic these naturally occurring processes creating a low input, self sufficient, perennial edible forest. The idea of "forest gardening" is just about 40 years old and has only recently entered America. It was originally established by Robert Hart in Europe, where, with virtually no GMO's, and significantly less pesticide and fertilizer use, the European commodity crop yields have kept up with, and actually surpassed those of the United States. A study, published by the International Journal of Agricultural Sustainability, comparing the United Nations Food and Agriculture Organization's data on American and European Maize production from 1961 to present, revealed that, in 1986 western Europe started producing slightly more than the US with 82899 hectograms/hectare and has remained in the lead ever since (see Figure 1 for trend data).

Figure 1: taken from Jack Heinemann's report: Sustainability and innovation in staple crop production in the US Midwest (2013)

Head of the study, Jack Heinemann at the University of Canterbury in New Zealand, further points out that "the difference between the estimated yield potential and actual yield or 'yield-gap' appears to be uniformly smaller in W. Europe than in the US Midwest." This means that we are not only producing less food than other more sustainable models, but we are using far more land and other resources to do so.

So how do agroforestry methods "double or triple crop yields while reducing the need for commercial fertilizers"? The principal focus is on creating a garden or farming system that mimics the structure and beneficial relationships one would find in a naturally occurring ecosystem—essentially designing agriculture to look like a forest. By doing so, the food production site ends up becoming a diverse polyculture that integrates trees, shrubs, perennial and annual plants, to maintain the sustainable resilience and productivity of a natural forest. The agroforestry process has two parts: remediating the soil using a process called sheet mulching, and planting perennial polycultures that will thrive within the healthier land. Initially, creating a small edible ecosystem that follows natural succession is a work intensive process; however once everything is on a roll, we can just sit back and watch nature take over, only intervening to harvest its produce, of course.

SHEET MULCHING: REMEDIATING THE SOIL

According to the 2011 FAO report, "25 percent of the world's food-producing soils are highly degraded or are rapidly being degraded" and if we also take into account moderately degraded soils "one-third of Earth's entire endowment of cropland is under threat." The poor quality of our soil directly contributes to productivity loss and requires many chemical inputs just to be able to support plant life. Sheet mulching, the first step of agroforestry, is process that returns organic matter, full of carbon and other vital nutrients to the soil, thereby rejuvenating overused and degraded land, and enhancing its fertility. This step is a cost effective and efficient way to transform any type of unproductive or unused land and is particularly important when a farmer must work with chemically treated grass. Sheet mulching essentially involves four layers: the preexisting base (most often, brown field or a lawn), a nitrogen rich compost layer, some type of weed barrier, and a carbon rich cover mulch. The following sheet mulching diagram (Figure 2) was taken from a report on the University of Massachusetts's on campus permaculture start up. Ryan Harb, head of the permaculture project, presented this diagram to show the four simple layers that he and fellow students used to remediate the soil on their 5000 square foot study site in 2009. This technique only cost them $0.20 per square foot and that is including $310 spent on supplementary soil minerals (which is optional).

Figure 2: Sheet Mulching Diagram

According to Harb, each layer can be made out of different materials depending on what is locally available. The nitrogen rich layer is ideally composed of finished compost, but any mix of organic material like manure, slashed vegetation or food waste will break down over time and suffice. The entire base is covered in about three to four inches of this compost, but it can additionally be put on top of the weed barrier for extra nutrients. Cardboard or newspaper is usually the cheapest weed barrier and can be overlapped in multiple sheets to completely seal the compost. The last carbon rich layer, often woodchips, leaves, store bought mulch or straw, helps to insulate and protect everything. Sheet mulching can be done in the spring or fall, and is a good way to keep the land productive throughout the winter because the layers are just left to sit for at least 6 months. This allows the organic materials to circulate and do their natural processes to revive the soil without the use of chemical fertilizers.

Studies show that "Converting the 64 million hectares of U.S. cropland currently planted in corn and soy- beans...would sequester 264 million tons of carbon dioxide; this is the equivalent of shutting down 207 (225- megawatt) coal-fired power plants, about 14 percent of the [America's] installed coal electric capacity." Not only would this conversion to sheet mulching be extremely helpful in mitigating climate change, but as students at Umass Amherst found out, the soil's carbon rich content and overall ecology directly influences crop health and makes for much more flavorful produce. High quality fruit will have a high brix score, which is an indicator for carbon levels. Boichar is another more traditional technique that can further increase the amount of carbon sequestered during sheet mulching. This process involves adding an additional layer of recycled charcoal into the mix because it retains huge amounts of carbon even years after burning. It is important to note however, that such use of charcoal is only beneficial if it is from a recycled source, seeing as it would waste just as much carbon to intentionally burn wood for fertilizing purposes, and thus the practice is not recommended for large scale farming.

Healthier soil due to sheet mulching has greater productive capacities. One of the best ways to measure improvements in soil health is to look at a soil profile before and after the process. Anyone get a soil profile done by sending a small amount of garden dirt to a lab, which will identify what nutrients it is in need of. When students at Umass Amherst did this initially, they found their soil to extremely low in "humus" or organic mater, as well as other nutrients like ammonia, phosphorus, potassium, calcium and magnesium. Rejuvenating this poor soil allowed the garden to produce roughly 1,000 pounds of fruits, vegetables, herbs and edible plants, just during the first season, and double this amount in the second year.

Another good tell for whether or not the soil is at its optimal level for production is to look at the weed patterns. Weed formation often occurs when the land is too dry or not densely packed and as Geoff Lawton, the managing director of The Permaculture Research Institute of Australia, words it, they function as a "volunteers" to help hold the soil together and prevent topsoil loss. The healthier the soil the fewer weeds, which means less labor input of weeding—sheet mulching not only reduces the need for synthetic fertilizers and nutrient inputs but also reduces the amount of labor wasted on weeding. From this perspective, the system is more productive and self-sustaining. Moreover, seeing as "herbicide-resistant crop technology has led to a 239 million kilogram (527 million pound) increase in herbicide use in the United States between 1996 and 2011," implementing sheet mulching techniques that result in less weeds would profoundly help the environment by lowering our heavy use of pesticides.

HUGELKULTUR AND KEYLINE FARMING

Alternatively, in permaculture several other productive processes, Hugelkultur and land Keyline, can be done during the sheet mulching cycle. These are both labor-intensive steps but they can result in more productive returns from the land. Hugelkultur is a means of minimizing water use and maximizing the land's water retention. It is a great option especially in cold climates because it extends the productive season of any site. Hugelkultur involves burying wood, full of carbon and other organic matter, deep below the ground. This wood is used to improve the earth's water retention capacity and will break down over time furthering the soil's biodiversity. To make it less work intensive one can also make " Hugelburms" which are basically raised flower beds made from simply stacking the wood on of the exposed ground layer and then preforming normal sheet mulching over them. This technique works just as well in terms of water retention, and also has the added benefit of enabling control over the shape of the land—a Keyline technique. Keyline farming is based on altering the contour line that forms with the slope of the land in order to adjust wind patterns and increase topsoil. The combination of Hugelkultur and Keyline farming allow for the most efficient use of water and the ability to slow water flow, maximizing its coverage by redirecting the flow of gravity all over the permaculture sight. This practice has most recently been used by Ben Falk, a homestead permaculture farmer in central VT, who integrated and used both Keyline and Hugelkultur to alter the contour of the land and work with the wind patterns.

Figure 4: Hugelkultur and Keyline working together

Figure 4 illustrates how Keyline farming can use two Hugelburms to create a swale in between where the wind is blocked and the ecosystem feels similar to that of a valley. Here and plants can flourish at high altitudes in temperate zones for a longer, moister season than usual.

PLANTING PERENNIAL POLYCULTURES AND GUILDS

The second phase of agroforestry is focused on a garden or farming system that mimics the structure and beneficial relationships one would find in a naturally occurring ecosystem—essentially designing agriculture to look like a forest. By doing so, the food production site ends up becoming a diverse polyculture that integrates trees, shrubs, perennial and annual plants, to maintain the sustainable resilience and productivity of a natural forest. The most manageable way to actually plan out where each plant will go is by selecting the tallest species to become a central element that the rest of the plant layers can be structured around. There will be more than one tall central species, each with multiple layers of diverse plants around it; these each become "guilds" or mutually beneficial miniature ecosystems, multiple of which make up an edible forest. Figure 5 presents a diagram of the multiple layers that should be present in each finalized guild. Certain plants play integral roles and benefit surrounding vegetation: soil improvers fix nitrogen levels, nectaries attract pollinating insects and prevent colony collapse, aromatic pest repellents, and ground coverage shrubs lock nutrients in place while keeping unwanted vegetation out.

Figure 5: Structural layers of agroforestry

With our soil loss nearing about 75 billion tons annually, the key to any form of agroforestry is including perennials and ground covers. Perennial plants, which are often the larger woody species such as fruit and nut trees, have much more time to establish themselves and therefore larger subterranean root structures. These roots hold the topsoil in place while also keeping the dirt at its optimal compactness, which lessens the need for weeds. Ground covers that hold year round are even more beneficial for the same reasons, since they simply cover more vulnerable soil. "Perennial farming systems are particularly suited to stabilizing slopes and preventing erosion on hillside farms" Writes Eric Toensmire, the author of "Perennial Vegetables" and teaching fellow for carbon sequestering agroforestry at Yale University. This is important since "roughly 45% of the world's farmland is classed as sloping at 8% or higher—regeneration of this quantity offarm1and with permanent agriculture would sequester 16.8 billion tons of carbon (at 25t/ha)." The woody nature of these species allows them to sequester more carbon in their bark. Additionally, large root structures can account for most of the plants biomass and 50% of its sequestration capacity. These woody roots are also tougher and can hold up better to insects and extreme weather conditions without requiring the same amount of maintenance and fertilizer as annuals. In fact, compared to conventional croplands, perennial permaculture methods produced 24-34 percent greater yields during drought season while using 28-32 percent less energy. This difference in productivity comes from the superior water retention capacity of healthy soils containing a high prevalence of carbon.

Considering the large size that perennial roots can achieve, it is important to take subterranean zoning into account when placing different plants next to each other. By thoroughly studying the different root patterns of plant species and the different elements of the soil profile that they each require, we have the ability to strategically place plants so that rather than competing for the same nutrients and the same depth in the earth, they are mutually beneficial. Not only do sustainable layouts like this allow for densely packed vegetation (and therefore higher food yields per unit of land), but they also result in a more natural niche that attracts ranges of bird and insect species—maintaining a full circle ecosystem that both feeds us, and helps to reverse our degradation of biodiverse land. It is on this basis that Geoff Lawton argues that we would need "about two to three percent of the present area that we use with industrial agriculture – two percent in equivalent area…to supply the world with the same nutrition that it presently requires, using permaculture principles."

COPPICE, INSTANT SUCCESSION, AND CARBON POTENTIAL

A prevalent worry with perennials is that they take too much time to establish and aren't financially productive enough within the first year or two. However, integration of perennials and annuals through practices like Coppice provides protection against soil erosion while also allowing for immediate productivity of annuals within a perennial edible forest. Coppice is an ancient no till technique that involves cutting down much of the woody plant growth (for firewood and renewed plant productivity), but still leaving these beneficial roots in place. During Coppice the soil can benefit from the changing nutrients involved in crop rotations, however only annual cover crops are rotated (often planted in between rows of perennials) while all the roots remain in the ground. This way the soil also still benefits from the carbon sequestration of perennial systems as "no-till" means that the organic mater will not be disturbed. Furthermore, timing of productivity is species dependent, and although the idea of agroforestry is to follow natural succession, we can actually accelerate this process. This idea is referred to by Eric Toensmire as "instant succession". Instant succession forest gardens are ones that allow us to create all the naturally occurring levels of succession, essentially all at once. Initially this method requires an intense investment of time and effort, but can result in a rapidly flourishing edible forest planted in dense patterns that should be self-sustaining without much maintenance for multiple years.

The exact amount of carbon sequestered varies drastically with humidity, location and implemented management. A properly designed permaculture sight, that uses not only the fundamentals of agroforestry polycultures, but also Hugelkultur, Keyline, Biochar and Coppice techniques, has an estimated potential of storing up to 228 tons (510720 pounds) of the world carbon in the land. And as Eric Toensmire points out, about 1 million tons (200 million pounds)/year are already sequestered just through the relatively small amount of agroforestry that exists globally.

THE NEED FOR RESEARCH, RECOGNITION, AND EDUCATION

This essay simply outlines the essentials of perennial agroforestry in hopes to entice others into researching and planting ecologically sustainable agriculture. There is still much to be uncovered about which other plant species and techniques can be beneficial in creating an interdependent web of edible forest structures. "Over the next few years "farming" has the potential to be widely recognized as an intellectually rigorous and rewarding field ready for more exploration." But for the spread of Permaculture to be possible we need much more scientific research and recognition. The social views around farming must shift to seeing it as an art of strategic planning and an intellectual science of designing ecosystems that mimic the structures in nature and maximize their interdependent benefits. This kind of cultivation of the land roots from a deep understanding of the science behind carbon emissions and soil ecology. We need people excited and enrolling in classes to learn about these techniques because these will be the people who are ready to actually manage a complex permaculture farm. For example, Caitlin Bergman and Dan McLeod, the two owners of Copia Farm in Johnstown, OH are extremely educated in progressive agriculture techniques. According to their website, "Dan received his Certificate in Permaculture Design from grasslands expert; Brock Doleman; certification in Keyline Design from world expert Darren Doherty; Advanced Permaculture Design Certification from Geoff Lawton; additional studies with Permaculture Co-Founder, David Holmgren; and Soil Advisor Certification from microbiologist Dr. Elaine Ingham." Meanwhile Caitlin spent over a decade as permaculture instructor at the Los Angeles Arboretum and Botanic Gardin, and then went on to organize and run her own permaculture design courses while becoming a professional soil sample analyst. As a result of years of research, in just one year the two were able to start their own 40-acre farm that incorporates all sorts of permaculture techniques, including Agroforestry, Coppice, Hugelkultur and Keyline (see figure 6) and begin making profit from a ton of produce.

Figure 6: Copia Design

Some progressive schools and universities are already taking the first steps by simply converting small patches of grass or unproductive land, into food producing micro ecosystems. These self-sufficient ecosystems are then used to educate students and the general public on the principals of permaculture and each individual's ability to cost effectively provide food while simultaneously benefiting the environment. Umass Amherst's Permaculture initiative is a fantastic example; they even won the White House's "Campus Champions of Change Challenge", got to present to Obama and appeared on MTV Act. However this all happened in March of 2012, and they are still the only public university in the US that has implemented a permaculture garden directly on campus and into their dining halls. So how is it possible that the average American has never even heard of permaculture? How come I spent two weeks researching Permaculture before I came across this White House Award? In my personal experience trying to research the permaculture topic using Yale University's library search engines, I struggled initially to find any of this information, even the articles published by Eric Toensmeier, an appointed professor at the Yale school of Forestry. The main barriers seem to be that relevant information either is not in scholarly articles but had a strong public profile (as was the case with Umass Amherst), or that it was in scholarly text and scientific/peer reviewed articles but only international ones that are not included in most searches. Other people are probably running into the same issue when they try to look into permaculture, plus there are so many relevant terms that obtaining a clear and concise understanding can be a challenge. But are these the real reasons for why permaculture has not made more progress, or is there an even larger issue?

Is it just the simple fact that most Americans are unaware of the problems inherent in industrialized agriculture? Since 2007 we have known that "89 percent of the technical mitigation potential is related to soil carbon sequestration, about 9 percent linked to mitigation of methane," thanks to the fourth assessment report of the Intergovernmental Panel on Climate Change (IPCC). Yet this information seems to have stimulated little to no change in the past seven years, considering that industrial agricultural methods are still surging onward. Even UN secretariat Ulrich Hoffman, in reference to our current agriculture paradigm, has stressed that: "As the world currently follows the worst case scenario of green house gas emissions projected by Intergovernmental Panel on Climate Change and International Energy Agency (i.e. implying a global warming of 4-6 degrees in Celsius) climate resilience and adaption, in combination with productivity increases should be prioritized." So how do we publicize that sectors heavy dependence on external mechanized and chemical inputs is exhausting the planets finite supply of fossil fuels, arable cropland, and uncontaminated water, while directly contributing to greenhouse gas emissions and therefore overall climate change?

CONCLUSION: A BOTTOM-UP PATH FORWARD

At the moment agroforestry is by no means a replacement for all the commercial fields of commodity crops, but unlike conventional farming, these permaculture techniques can be applied almost anywhere, on small and large scales. This means that any patch of unproductive grass can be converted into a thriving, food producing micro ecosystem. Such forest gardens could be in every park, making up the front yards of churches, town halls, and eventually individual homes. As Eric Toensmire puts it, "Permanent agriculture needs to be spread much, much more widely if it is to have a significant impact." Sustainable food production through agroforestry techniques provides a means for nature to reclaim lost land one patch of chemically treated, mechanically watered, constantly mowed grass at a time—starting on a small scale and spreading all over both urban and rural locations. Eventually the resulting "bottom up" approach to changing our food system makes each community more locally self-sufficient with the potential to lessen America's reliance on large-scale unsustainable agribusinesses. But the first, and much bigger step for America is recognizing that land is lost; that just because we see green grass, dump a bunch of fertilizer on it, and have a supermarket full of food, does not mean our system is at all flawless. Somehow we have to overcome this common inertia towards pressing environmental issues and begin to implement existing knowledge.

REFERENCES

1. From Agriculture to Permaculture (State of the World 2010: Transforming Cultures from Consumerism to Sustainability, 47-53). Albert Bates and Toby Hemenway. 2010. http://blogs.worldwatch.org/transformingcultures/wp-content/uploads/2010/07/From-Agriculture-to-Permaculture-Bates-and-Hemenway.pdf
2. Impacts of Genetically Engineered Crops on Pesticide Use in the U.S. (Environmental Sciences Europe, vol. 24). Charles M. Benbrook. 2012. https://doi.org/10.1186/2190-4715-24-24
3. After the Green Revolution: Sustainable Agriculture for Development (Routledge). Gordon R. Conway and Edward B. Barbier. 2013.
4. Copia: A Farm Designed with Nature: The Farmers (Copia Farm website). 2014. http://www.copiaohio.com/#!the-farmers/c24i2
5. Perennial Crops for Food Security: Proceedings of the FAO Expert Workshop (Biodiversity & Ecosystem Services in Agricultural Production Systems, Rome, Italy, 1-409). Stan Cox, Caterina Batello, and Len Wade. 2014. http://www.fao.org/3/a-i3495e.pdf
6. Introduction to Introduction to Food Sovereignty: Food and Democracy (8-13). Marcin Gerwin (introduction; book by Rob Avis and Sue Branford). 2011.
7. Beyond Sustainability Homes: Pilot Project in Amherst MA (beyondsustain.org). Ryan Harb. http://www.beyondsustain.org/food-production.html
8. Lawns to Gardens: Growing Your Own Food for Economic and Environmental Savings (Master's thesis, University of Massachusetts Amherst). Ryan Harb. 2010. http://www.beyondsustain.org/uploads/6/9/5/3/6953085/ryan_harb_-_yard_-transformation_-_a_how-to_guide-libre.pdf
9. Chapter 3: What Will We Eat as the Oil Runs Out? (in Introduction to Food Sovereignty: Food and Democracy, ed. Marcin Gerwin, 31-46). Richard Heinberg. 2011.
10. Sustainability and Innovation in Staple Crop Production in the US Midwest (International Journal of Agricultural Sustainability, vol. 12, 71-88). Jack A. Heinemann, Melanie Massaro, Dorien S. Coray, and Jiajun Dale Wen. 2013. https://doi.org/10.1080/14735903.2013.806408
11. Chapter 12: Seeing the Garden in the Jungle (in Introduction to Food Sovereignty: Food and Democracy, ed. Marcin Gerwin, 104-08). Toby Hemenway. 2011.
12. Required Fundamental Transformation of Agriculture (UNCTAD Trade and Environment Review 2013: Wake Up Before It Is Too Late, 20-28). Ulrich Hoffman. 2013. http://unctad.org/en/PublicationsLibrary/ditcted2012d3_en.pdf
13. Chapter 13 Edible Forest Gardens: An Invitation to Adventure (in Food and Democracy: Introduction to Food Sovereignty, ed. Marcin Gerwin, 109-20; previously published in Edible Forest Gardens: A Delicious and Practical Ecology, rev. ed., vol. 1). David Jake (Jacke). 2011.

14. UMass Amherst Permaculture Project Wins White House Award (Boston Globe Media Partners). Tracy Jan. 2012. http://www.boston.com/2012/03/13/champion/r2XazdAk-dr4SrYpzQWBT7K/story.html
15. Permaculture: Your Way to Sustainable Living (in Food and Democracy: Introduction to Food Sovereignty, ed. Marcin Gerwin, 127-32; excerpt from Veritas Magazine, August 2010). Geoff Lawton. 2011.
16. Orchestrating Famine (in Food and Democracy: Introduction to Food Sovereignty, ed. Marcin Gerwin, 15-22; excerpt from Permaculture Research Institute of Australia, 2008). Craig Mackintosh. 2011.
17. Sustainable Organizing: A Multiparadigm Perspective of Organizational Development and Permaculture Gardening (Journal of Leadership and Organizational Studies, 355-68). Delia Mannen. 2012.
18. Reaping the Benefits: Science and the Sustainable Intensification of Global Agriculture (The Royal Society). The Royal Society. 2009. https://royalsociety.org/~/media/Royal_Society_Content/policy/publications/2009/4294967719.pdf
19. The Resilient Farm and Homestead — Growing Rice in a Cold Climate, Is This Possible? (Permaculture Research Institute of Australia, produced by Geoff Lawton). Ben Falk (performer); Geoff Lawton (producer). 2013. http://www.geofflawton.com/fe/60662-cold-climate-permaculture
20. Global Hunger Crisis: Tackling Food Insecurity in Developing Countries (Pluto Press). Majda Bne Saad. 2013. http://site.ebrary.com/lib/yale/detail.action?docID=10675215
21. Addressing the Problem Not Just the Symptoms: Climate Stability with Permanent Agriculture (Permaculture Activist, no. 80, 22-28). Eric Toensmeier. 2011. http://www.perennialsolutions.org/wp-content/uploads/2011/07/Climate-Stability-with-Perm-Ag_Toensmeier_2011.pdf
22. Carbon Farming 5: Perennial Farming Systems (lecture, Northeast Organic Farming Association (NOFA) Summer Conference; YouTube video). Eric Toensmeier. 2011. https://www.youtube.com/watch?v=QwKzbNvDrhY
23. Perennial Agriculture Now: A Global Review of Existing Perennial Staple Crops by Climate Region and Nutrient Profile (lecture, American Society of Agronomy Annual Meeting; YouTube video). Eric Toensmeier and Rafter Ferguson. 2014. https://www.youtube.com/watch?v=em9sdBX0T_A